

FINTRUST BANK

Power BI Dashboard Build Guide

Data Analytics Track | AnalystLab Africa Experience Lab

Field	Detail
Document	Power BI Dashboard Build Guide
Purpose	Reproducible step-by-step instructions and ready-to-paste DAX so the .pbix can be built in Power BI Desktop
Environment limitation	Power BI Desktop is a Windows-only desktop application and could not be run in the Linux sandbox used to prepare this project. This guide, together with the published interactive HTML dashboard, stands in for the .pbix until it is built in the actual application.
Companion file	FinTrust_Week2_Analytics_Dashboard.html — a working interactive substitute built from the same cleaned data, with independent Customer Segment and Channel slicers and 9 live charts
Data source	FinTrust_Customer_Data_Clean.csv, FinTrust_Transaction_Data_Clean.csv
Version	1.0 — 19 September 2026

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1. Why This Guide Exists

The Week 2 assignment asks for `FinTrust_Week2_Analytics_Dashboard.pbix`, built in Microsoft Power BI Desktop. Power BI Desktop is Windows-only, proprietary software with no command-line or Linux equivalent, so it could not be installed or run in the sandboxed environment used to prepare the rest of this Week 2 submission.

Two things stand in for the missing .pbix so the work is verifiable rather than simply asserted:

- This guide — every step, every visual, and every DAX measure needed to reproduce the dashboard exactly, in the order Power BI expects them.
- `FinTrust_Week2_Analytics_Dashboard.html` — a working, interactive dashboard built from the same cleaned data, with independent Customer Segment and Channel slicers and nine charts, published as a functioning substitute you can open and click through today.

Building the actual .pbix from this guide should take under 30 minutes in Power BI Desktop, since every measure, column and chart choice has already been decided and tested against the real data.

2. Get the Data In

1. Open Power BI Desktop → Home → Get Data → Text/CSV.
2. Load `FinTrust_Customer_Data_Clean.csv`. In the preview window, confirm Power BI has detected `Customer_ID` as Text and `Age/Tenure_Months` as Whole Number, then click Transform Data (not Load) so both tables land in Power Query first.
3. Repeat Get Data → Text/CSV for `FinTrust_Transaction_Data_Clean.csv`.
4. In Power Query, confirm `Transaction_DateTime` is typed as Date/Time (Power BI usually infers this correctly from the CSV) and `Amount_NGN` as Decimal Number.
5. Click Close & Apply.

Both CSVs are already cleaned (nulls imputed, date/amount bands added, ambiguous labels disambiguated) — no further Power Query transformation is required before modelling.

3. Build the Data Model

3.1 Relationship

6. Go to the Model view.
7. Power BI should auto-detect a relationship on `Customer_ID`. If not, drag `Customer_ID` from the Transaction table onto `Customer_ID` in the Customer table.
8. Set the relationship: Customer (1) → Transaction (*), single direction, filtering from Customer to Transaction.

3.2 Date table

The transaction CSV already carries Year, Quarter, Month, Week, `Day_Of_Week` and Hour as columns (added in Part A cleaning), so a separate Date dimension is optional for this build. For a production model, mark a proper Date table and set `Transaction_DateTime` as its Mark as Date Table source.

4. DAX Measures

Create a dedicated measures table first (Model view → Enter Data → name it "_Measures", no columns needed) and add every measure below to it, so they are not scattered across the physical tables.

Measure name	DAX
Total Transactions	Total Transactions = DISTINCTCOUNT(Transaction[Transaction_ID])
Total Value	Total Value = SUM(Transaction[Amount_NGN])
Avg Ticket	Avg Ticket = DIVIDE([Total Value], [Total Transactions])
Median Ticket	Median Ticket = MEDIAN(Transaction[Amount_NGN])
Success Rate	Success Rate = DIVIDE(CALCULATE([Total Transactions], Transaction[Transaction_Status]="Successful"), [Total Transactions])
Failure Rate	Failure Rate = DIVIDE(CALCULATE([Total Transactions], Transaction[Transaction_Status] IN {"Failed","Reversed","Pending"}), [Total Transactions])
Failed Reversed Value	Failed Reversed Value = CALCULATE([Total Value], Transaction[Transaction_Status] IN {"Failed","Reversed"})
Risk Review Rate	Risk Review Rate = DIVIDE(CALCULATE([Total Transactions], Transaction[Risk_Review_Flag]="Yes"), [Total Transactions])
Value Under Review	Value Under Review = CALCULATE([Total Value], Transaction[Risk_Review_Flag]="Yes")
Digital Share	Digital Share = DIVIDE(CALCULATE([Total Transactions], Transaction[Channel] IN {"Mobile App","Web","USSD"}), [Total Transactions])
Total Customers	Total Customers = DISTINCTCOUNT(Customer[Customer_ID])
Value Per Customer	Value Per Customer = DIVIDE([Total Value], [Total Customers])
P90 Amount	P90 Amount = PERCENTILE.INC(Transaction[Amount_NGN], 0.9)
High Value Risk Rate	High Value Risk Rate = CALCULATE([Risk Review Rate], Transaction[Amount_NGN] >= [P90 Amount])
Intl Risk Rate	Intl Risk Rate = CALCULATE([Risk Review Rate], Transaction[International_Transaction]="Yes")

5. Report Pages and Visuals

Build four pages, matching the Week 1 Dashboard Wireframe exactly so the two documents stay in sync.

Page 1 — Executive Overview

- Card visuals: Total Value, Total Transactions, Median Ticket (with Avg Ticket as a secondary card), Success Rate, Digital Share.
- Line chart: Transaction_DateTime (by day) on axis, Total Value on values.
- Clustered bar chart: Transaction_Type on axis, Total Value on values, sorted descending.
- Combo chart: Channel on axis, Total Value as columns, Total Transactions as a line on secondary axis.
- Bar chart: Customer_Segment on axis, Total Value on values — do not sort by an implied hierarchy.
- Bar chart: City on axis, Total Value on values, sorted descending (no map visual — see Section 6).

Page 2 — Channel & Operations

- Card visuals: Success Rate, Failure Rate, Failed Reversed Value, Digital Share.
- 100% stacked bar chart: Channel on axis, Transaction_Status as legend, Total Transactions on values, "% of grand total" data label.
- Scatter chart: Total Transactions on X, Failure Rate on Y, Channel as legend/details, bubble size = Total Value.
- Bar chart: Device_Type on axis, Total Transactions on values only — do not add Failure Rate or Risk Review Rate to this chart or its tooltip.
- Matrix visual: Channel on rows, Transaction_Type on columns, Failure Rate on values with conditional formatting (colour scale).

Page 3 — Risk Monitoring

- Text box (locked, always visible): "Risk_Review_Flag is a synthetic educational label indicating a transaction selected for review. Not a fraud determination."
- Card visuals: Risk Review Rate, Value Under Review, International share, count of flagged transactions.
- Bar chart: International_Transaction on axis, Risk Review Rate on values, with a constant reference line at the book-wide Risk Review Rate.
- Column chart: Amount decile (bin Amount_NGN into 10 equal buckets, or use a calculated column) on axis, Risk Review Rate on values.
- Line chart: Hour on axis, Risk Review Rate on values, with the 00:00-05:59 band highlighted using a shape or a conditional-format reference band.
- Table visual: Transaction_ID, Transaction_DateTime, Amount_NGN, Location, Transaction_Status, filtered to Risk_Review_Flag = "Yes" — set as the drill-through target page for every other visual on this page.

Page 4 — Customer 360

- Card visuals: Value Per Customer, average transactions per customer, Active rate, average Digital_Engagement_Score.
- Matrix visual: Customer_Segment on rows, Account_Type on columns, count of Customer_ID on values.
- Stacked bar chart: Customer_Segment on axis, Account_Status as legend, count of Customer_ID on values.
- Scatter chart: Digital_Engagement_Score on X, count of transactions per customer on Y, Customer_Segment as legend.
- Bar chart: Monthly_Income_Band on axis (sorted using the Monthly_Income_Band_Sort column via Column tools → Sort by column), Value Per Customer on values.
- Histogram (bar chart on Tenure_Band): Tenure_Band on axis, count of Customer_ID on values.

6. Slicers and Filters

- Add a slicer panel on every page: Date range (Transaction_DateTime), Customer_Segment, Channel, City.

- Set slicer sync (View → Sync Slicers) for Date range and Customer_Segment across all four pages, so a filter set on Page 1 carries to Pages 2-4 consistently.
- Do not add a filled map visual anywhere in the report. Transaction Location is statistically independent of the customer's home City (chi-square $p = 0.541$, see the Week 1 Data Quality Assessment Report) — a map would visually imply a geographic pattern the data does not contain. Use the sorted bar charts specified above instead.

7. Formatting Checklist

- Apply a consistent theme: View → Themes → Customize current theme, using the accent colour #1B5E9C to match the Week 1 wireframe and this dashboard.
- Every card showing Avg Ticket must be paired with the Median Ticket card next to it — the distribution is skewed 4.5x (see the EDA notebook, Section 4) and the mean alone is misleading.
- Format Total Value and Value Under Review as currency with the ₦ (NGN) symbol.
- Add a text box on every page footer: "Risk_Review_Flag is a synthetic educational label. All data are fictional, AnalystLab Africa training purposes."
- Publish or save as FinTrust_Week2_Analytics_Dashboard.pbix in the Week 2 submission folder.

8. Validation — Check Your Build Against These Numbers

After building, cross-check every card on Page 1 against the table below. These are the actual, computed values from the cleaned data (same figures used in the SQL analysis, the EDA notebook, and the published HTML dashboard) — if your Power BI card disagrees, the model or a filter is wrong, not the target.

Measure	Expected value (Q1 2026, unfiltered)
Total Customers	1,500
Total Transactions	12,000
Total Value	₦560,477,355
Avg Ticket	₦46,706
Median Ticket	₦10,306
Success Rate	90.47%
Failure Rate	9.53%
Risk Review Rate	19.60%
Value Under Review	₦161,783,631
Digital Share	65.50%
Intl Risk Rate	36.88% (vs 18.88% domestic)

End of guide. All data referenced are fictional and synthetic, created for AnalystLab Africa training purposes.